

DARKE COUNTY AP, OH, USA

WMO: 720713

Lat: 40.204N

Lon: 84.532W

Elev: 307

StdP: 97.69

Time Zone: -5.00 (NAE)

Period: 10-19

WBAN: 00251

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.0	-20.9	0.6	-16.9	-17.5	0.8	-13.6	12.6	-3.0	11.5	-3.4	2.7	280	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.2	22.4	31.0	22.2	29.9	21.7	25.2	28.9	24.3	28.1	23.4	27.1	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.6	27.2	22.9	18.3	26.0	22.4	17.8	25.3	79.2	28.4	74.7	27.8	71.5	27.3	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	8.9	7.9	-21.0	34.1	3.8	2.0	-23.7	35.6	-25.9	36.8	-28.1	37.9	-30.8	39.4		
(5)			-21.3	26.7	3.8	0.4	-24.0	27.0	-26.2	27.3	-28.4	27.5	-31.1	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-2.9	0.0	3.8	10.6	17.3	21.3	22.8	21.5	18.6	12.2	4.7	0.4		
	DBStd	10.31	6.77	6.47	6.06	4.86	5.04	3.33	3.02	2.69	4.31	5.17	5.14	5.55		
	HDD10.0	1463	402	286	212	49	6	0	0	0	0	36	172	299		
	HDD18.3	3174	659	514	451	233	81	12	3	6	51	202	408	555		
	CDD10.0	1796	1	6	19	69	232	339	396	356	259	103	13	3		
	CDD18.3	466	0	0	1	3	48	102	141	103	59	11	0	0		
	CDH23.3	4057	0	0	6	38	418	847	1222	850	574	99	2	0		
	CDH26.7	1236	0	0	0	2	109	263	407	227	207	21	0	0		
(14) Wind	WSAvg	3.3	4.4	4.2	4.1	4.1	3.3	3.0	2.0	1.8	2.2	3.2	3.7	3.8		
(15) Precipitation	PrecAvg	1009	73	54	76	97	111	113	109	88	71	72	75	74		
	PrecMax	1305	233	117	163	180	192	198	241	148	166	160	169	213		
	PrecMin	771	23	19	25	37	40	64	25	27	18	13	25	25		
	PrecStd	147	45	26	37	38	40	39	51	33	37	36	36	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	18.6	25.0	27.1	32.0	32.8	33.8	32.3	33.0	29.1	22.0	16.1		
		MCWB	13.7	13.3	16.9	16.6	20.6	22.0	23.5	22.7	21.7	19.1	16.5	14.5		
	2%	DB	12.1	15.2	20.0	24.1	29.7	31.2	32.1	30.9	31.3	26.2	17.8	12.9		
		MCWB	11.1	13.2	14.6	15.5	20.0	21.6	23.8	22.7	21.1	17.7	13.4	12.0		
	5%	DB	9.1	12.5	16.3	22.1	27.6	29.9	30.9	29.1	29.1	22.9	15.9	11.1		
		MCWB	8.3	11.0	12.8	14.6	19.2	20.8	23.2	22.4	20.7	16.9	11.9	10.4		
	10%	DB	6.4	9.1	13.0	19.1	26.0	27.7	29.0	27.6	27.1	21.1	12.7	7.7		
		MCWB	5.0	7.1	10.3	13.5	18.5	20.2	22.6	21.3	19.2	16.3	9.7	6.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	15.5	17.2	19.1	22.4	25.1	26.6	25.9	24.5	22.1	17.3	15.3		
		MCDB	15.5	17.6	22.5	24.5	26.5	29.6	29.6	28.8	28.7	25.9	19.0	15.8		
	2%	WB	11.2	13.1	15.4	17.2	21.6	23.9	25.6	24.9	23.3	20.3	15.0	12.5		
		MCDB	11.5	13.8	19.4	20.8	26.0	27.9	29.1	28.1	27.9	23.1	17.5	13.3		
	5%	WB	8.0	10.9	13.4	16.0	20.7	22.9	24.8	23.8	22.3	18.7	12.3	10.1		
		MCDB	8.6	13.1	15.8	19.4	25.1	26.9	28.6	27.1	26.3	22.1	14.6	10.9		
	10%	WB	5.2	7.2	11.3	14.8	19.9	22.0	23.7	22.8	21.2	17.0	10.1	7.1		
		MCDB	6.3	8.5	12.9	18.5	24.7	25.8	27.5	25.9	24.6	20.0	12.9	7.6		
(35) Mean Daily Temperature Range	MDBR		7.7	8.2	8.8	11.7	11.0	10.8	10.8	11.7	12.5	11.3	9.0	6.6		
	5% DB	MCDBR	9.9	12.0	12.3	15.1	13.0	13.0	11.8	13.0	14.8	14.1	13.1	9.6		
		MCWBR	9.0	9.8	8.4	8.2	5.4	5.1	5.1	6.0	5.5	6.9	8.6	8.7		
	5% WB	MCDBR	9.4	11.5	11.0	12.0	10.8	10.0	9.9	10.1	12.1	11.1	11.3	9.2		
		MCWBR	9.3	10.1	8.0	8.5	5.3	5.2	5.5	5.6	5.6	6.5	8.8	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.320	0.344	0.400	0.441	0.442	0.446	0.440	0.399	0.354	0.324	0.299		
	taud		2.418	2.351	2.363	2.218	2.146	2.197	2.198	2.198	2.285	2.404	2.445	2.455		
	Ebn at Noon		863	893	909	873	841	839	831	825	837	843	827	837		
	Edh at Noon		81	100	111	137	151	144	142	139	119	95	79	73		
(44) All-Sky Solar Radiation	RadAvg		1.67	2.57	3.56	4.76	5.47	6.07	6.02	5.47	4.45	3.08	2.04	1.40		
	RadStd		0.15	0.28	0.30	0.42	0.46	0.43	0.34	0.26	0.39	0.28	0.25	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	N/A

Nomenclature: See separate page